Math Notes Blueprint

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Chapter 1

Mathematics

1.1 Linear Algebra

1.1.1 Structural Linear Algebra

Vector Spaces and Linear Structure

The guiding idea is that a vector space is not a list of coordinates first. It is a space of objects that can be added and scaled; subspaces, spans, independence, bases, and dimension describe how such spaces are generated, while direct sums and quotient spaces describe how they split apart or collapse selected directions.

Definition 1.1.1.1. *Vector Space*
Let $\mathbb{F}$ be a field. A vector space over $\mathbb{F}$ is a set $\mathcal{V}$ together with operations of vector addition and scalar multiplication such that, for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathcal{V}$ and all $a, b \in \mathbb{F}$, the following axioms hold:

i. vector addition is commutative: $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$;

ii. vector addition is associative: $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$;

iii. there exists an additive identity $\mathbf{0} \in \mathcal{V}$ such that $\mathbf{0} + \mathbf{v} = \mathbf{v}$;

iv. for each $\mathbf{v} \in \mathcal{V}$ there exists an additive inverse $-\mathbf{v} \in \mathcal{V}$ such that $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$;

v. scalar multiplication has identity: $1\mathbf{v} = \mathbf{v}$;

vi. scalar multiplication is compatible with multiplication in $\mathbb{F}$: $(ab)\mathbf{v} = a(b\mathbf{v})$;

vii. scalar multiplication distributes over vector addition: $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$;

viii. scalar multiplication distributes over addition in $\mathbb{F}$: $(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$.

If $\mathcal{V} = \{\mathbf{0}\}$, then $\mathcal{V}$ is a *zero vector space*; if $\mathcal{V} \neq \{\mathbf{0}\}$, then $\mathcal{V}$ is a *nonzero vector space*.

Definition 1.1.1.2. *Finitely Nonzero Sequence Space*
Let $\mathbb{F}^{(\mathbb{N})}$ denote the set of all sequences $(x_1, x_2, \ldots)$ of elements of $\mathbb{F}$ such that $x_j = 0$ for all but finitely many $j \in \mathbb{N}$. Addition and scalar multiplication are defined componentwise.

Proposition 1.1.1.3. *Finitely Nonzero Sequences Form a Vector Space*
The set $\mathbb{F}^{(\mathbb{N})}$ is a vector space over $\mathbb{F}$ under componentwise addition and scalar multiplication.

Proof. The zero sequence belongs to $\mathbb{F}^{(\mathbb{N})}$. If $\mathbf{x} = (x_1, x_2, \ldots)$ and $\mathbf{y} = (y_1, y_2, \ldots)$ are in $\mathbb{F}^{(\mathbb{N})}$ and $a \in \mathbb{F}$, then the nonzero entries of $a\mathbf{x} + \mathbf{y}$ can occur only among the finitely many indices where $\mathbf{x}$ or $\mathbf{y}$ is nonzero. Thus $a\mathbf{x}+\mathbf{y} \in \mathbb{F}^{(\mathbb{N})}$, and the vector space axioms are inherited componentwise from $\mathbb{F}$.
Structural Commentary: Finite support is preserved by finite linear combinations.
Key Insight: Componentwise operations cannot create infinitely many new nonzero entries. □

Theorem 1.1.1.4. *Basic Consequences of Vector Space Axioms*
Let $\mathcal{V}$ be a vector space over $\mathbb{F}$. Then the additive identity and additive inverses are unique. Moreover, for all $a \in \mathbb{F}$ and $\mathbf{v} \in \mathcal{V}$, $0\mathbf{v} = \mathbf{0}$, $a\mathbf{0} = \mathbf{0}$, and $a\mathbf{v} = \mathbf{0}$ if and only if $a = 0$ or $\mathbf{v} = \mathbf{0}$.

Proof. If $\mathbf{0}$ and $\mathbf{0}'$ are additive identities, then $\mathbf{0} = \mathbf{0}+\mathbf{0}' = \mathbf{0}'$, so the additive identity is unique. If $\mathbf{w}$ and $\mathbf{w}'$ are additive inverses of $\mathbf{v}$, then $\mathbf{w} = \mathbf{w}+\mathbf{0} = \mathbf{w}+(\mathbf{v}+\mathbf{w}') = (\mathbf{w}+\mathbf{v})+\mathbf{w}' = \mathbf{0}+\mathbf{w}' = \mathbf{w}'$, so additive inverses are unique. Since $0\mathbf{v} = (0+0)\mathbf{v} = 0\mathbf{v} + 0\mathbf{v}$, cancellation gives $0\mathbf{v} = \mathbf{0}$.
Similarly, $a\mathbf{0} = a(\mathbf{0}+\mathbf{0}) = a\mathbf{0} + a\mathbf{0}$, so $a\mathbf{0} = \mathbf{0}$. If $a = 0$ or $\mathbf{v} = \mathbf{0}$, then the preceding identities give $a\mathbf{v} = \mathbf{0}$. Conversely, suppose $a\mathbf{v} = \mathbf{0}$ and $a \neq 0$. Then a^{-1} exists in $\mathbb{F}$, and $\mathbf{v} = 1\mathbf{v} = (a^{-1}a)\mathbf{v} = a^{-1}(a\mathbf{v}) = a^{-1}\mathbf{0} = \mathbf{0}$.
Structural Commentary: These identities are the basic algebraic hygiene of vector spaces. They let later arguments use cancellation, zero vectors, and inverses without reopening the axioms each time.
Key Insight: The vector space axioms force the zero and inverse behaviour to be rigid, not optional structure. □

Corollary 1.1.1.5. *Let $\mathcal{V}$ be a vector space over $\mathbb{F}$. Then $(-1)\mathbf{v} = -\mathbf{v}$ for every $\mathbf{v} \in \mathcal{V}$.*

Proof. Let $\mathbf{v} \in \mathcal{V}$. We show that $(-1)\mathbf{v} + \mathbf{v} = \mathbf{0}$. By the vector space axioms and Theorem 1.1.1.4,

$$(-1)\mathbf{v} + \mathbf{v} = (-1)\mathbf{v} + 1\mathbf{v} = (-1+1)\mathbf{v} = 0\mathbf{v} = \mathbf{0}$$

Thus $(-1)\mathbf{v}$ is the additive inverse of $\mathbf{v}$, so $(-1)\mathbf{v} = -\mathbf{v}$.
Structural Commentary: Scalar multiplication by -1 is exactly the same operation as taking an additive inverse.
Key Insight: Negation is not an extra operation; it is already encoded by the field action. □

Definition 1.1.1.6. A *subspace* of a vector space $\mathcal{V}$ over $\mathbb{F}$ is a subset $\mathcal{U} \subseteq \mathcal{V}$ that is itself a vector space over $\mathbb{F}$ with the same vector addition and scalar multiplication as in $\mathcal{V}$.

Theorem 1.1.1.7. *Subspace Test*
Let $\mathcal{V}$ be a vector space over $\mathbb{F}$ and let $\mathcal{U}$ be a nonempty subset of $\mathcal{V}$. Then $\mathcal{U}$ is a subspace of $\mathcal{V}$ if and only if $c\mathbf{u} + \mathbf{v} \in \mathcal{U}$ for all $\mathbf{u}, \mathbf{v} \in \mathcal{U}$ and all $c \in \mathbb{F}$.

Proof. ($\Rightarrow$) If $\mathcal{U}$ is a subspace, then $c\mathbf{u}+\mathbf{v} \in \mathcal{U}$ for all $\mathbf{u}, \mathbf{v} \in \mathcal{U}$ and all $c \in \mathbb{F}$ because $\mathcal{U}$ is closed under scalar multiplication and vector addition.

($\Leftarrow$) Suppose that $c\mathbf{u} + \mathbf{v} \in \mathcal{U}$ whenever $\mathbf{u}, \mathbf{v} \in \mathcal{U}$ and $c \in \mathbb{F}$. We verify that $\mathcal{U}$ is a vector space with the operations inherited from $\mathcal{V}$. Since $\mathcal{U}$ is nonempty, let $\mathbf{u} \in \mathcal{U}$. By Corollary 1.1.1.5, $\mathbf{0} = (-1)\mathbf{u} + \mathbf{u} \in \mathcal{U}$. If $\mathbf{u} \in \mathcal{U}$, then $-\mathbf{u} = (-1)\mathbf{u} + \mathbf{0} \in \mathcal{U}$. If $\mathbf{u}, \mathbf{v} \in \mathcal{U}$ and $c \in \mathbb{F}$, then $c\mathbf{u} = c\mathbf{u} + \mathbf{0} \in \mathcal{U}$ and $\mathbf{u} + \mathbf{v} = 1\mathbf{u} + \mathbf{v} \in \mathcal{U}$. Thus $\mathcal{U}$ contains $\mathbf{0}$ and is closed under additive inverses, scalar multiplication, and vector addition; the remaining vector space axioms are inherited from $\mathcal{V}$. Hence $\mathcal{U}$ is a subspace.

Structural Commentary: The theorem compresses the subspace axioms into one closure condition. This is the practical test used whenever a subset is proposed as a linear space.

Key Insight: To prove a subset is a subspace, prove it survives one affine-looking linear combination. □

Proposition 1.1.1.8. *Intersection of Subspaces*
Let $\mathcal{U}$ and $\mathcal{W}$ be subspaces of $\mathcal{V}$. Then $\mathcal{U} \cap \mathcal{W} = \{\mathbf{v} : \mathbf{v} \in \mathcal{U} \text{ and } \mathbf{v} \in \mathcal{W}\}$ is a subspace of $\mathcal{V}$.

Proof. Since $\mathbf{0} \in \mathcal{U}$ and $\mathbf{0} \in \mathcal{W}$, we have $\mathbf{0} \in \mathcal{U} \cap \mathcal{W}$, so the intersection is nonempty. If $\mathbf{u}, \mathbf{v} \in \mathcal{U} \cap \mathcal{W}$ and $c \in \mathbb{F}$, then $c\mathbf{u} + \mathbf{v}$ lies in both $\mathcal{U}$ and $\mathcal{W}$ because both are subspaces. Hence $c\mathbf{u} + \mathbf{v} \in \mathcal{U} \cap \mathcal{W}$, and Theorem 1.1.1.7 shows that $\mathcal{U} \cap \mathcal{W}$ is a subspace of $\mathcal{V}$.
Structural Commentary: Intersections preserve every closure condition shared by the subspaces. The subspace test packages this into one check.
Key Insight: Common vectors of two subspaces still form a subspace because the shared part remains closed under linear combinations. □

Definition 1.1.1.9. *Lists, Linear Combinations, and Span*
A *list* of vectors in $\mathcal{V}$ is a finite ordered sequence $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$ in $\mathcal{V}$; the empty list is denoted $()$.
If $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$ is nonempty and $a_1, \ldots, a_m \in \mathbb{F}$, then $a_1\mathbf{v}_1 + \cdots + a_m\mathbf{v}_m$ is a *linear combination* of β.
For $E \subseteq \mathcal{V}$, the *span* of E is $\operatorname{span}(E) := \{a_1\mathbf{v}_1 + \cdots + a_m\mathbf{v}_m : m \in \mathbb{N},\ a_j \in \mathbb{F},\ \mathbf{v}_j \in E\}$. $\operatorname{span}(\emptyset) = \{\mathbf{0}\}$.
If $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$ is a nonempty list, then $\operatorname{span}(\beta) = \operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_m) := \operatorname{span}(\{\mathbf{v}_1, \ldots, \mathbf{v}_m\})$.
A subset $E \subseteq \mathcal{V}$ *spans* $\mathcal{V}$, or is a *spanning set* for $\mathcal{V}$, if $\operatorname{span}(E) = \mathcal{V}$.
A list β in $\mathcal{V}$ *spans* $\mathcal{V}$, or is a *spanning list* for $\mathcal{V}$, if $\operatorname{span}(\beta) = \mathcal{V}$.

Theorem 1.1.1.10. *Span is the Smallest Containing Subspace*
Let $E \subseteq \mathcal{V}$. Then $\operatorname{span}(E)$ is the smallest subspace of $\mathcal{V}$ containing E.

Proof. If $E = \emptyset$, then $\operatorname{span}(E) = \{\mathbf{0}\}$, which is a subspace. Suppose $E \neq \emptyset$. If $\mathbf{x}, \mathbf{y} \in \operatorname{span}(E)$ and $c \in \mathbb{F}$, then $\mathbf{x}$ and $\mathbf{y}$ are finite linear combinations of vectors in E. Hence $c\mathbf{x} + \mathbf{y}$ is also a finite linear combination of vectors in E, so $c\mathbf{x} + \mathbf{y} \in \operatorname{span}(E)$. By Theorem 1.1.1.7, $\operatorname{span}(E)$ is a subspace of $\mathcal{V}$.
Each $\mathbf{v} \in E$ satisfies $\mathbf{v} = 1\mathbf{v} \in \operatorname{span}(E)$, so $E \subseteq \operatorname{span}(E)$. If $\mathcal{U}$ is a subspace of $\mathcal{V}$ with $E \subseteq \mathcal{U}$, then closure under vector addition and scalar multiplication forces every finite linear combination of vectors in E to lie in $\mathcal{U}$. Thus $\operatorname{span}(E) \subseteq \mathcal{U}$. Therefore $\operatorname{span}(E)$ is the smallest subspace of $\mathcal{V}$ containing E.
Structural Commentary: Span takes a set of vectors and closes it under all finite linear combinations. The result is precisely the smallest subspace forced by those vectors.
Key Insight: Nothing beyond linear closure is added. Every containing subspace must already contain span. □

Proposition 1.1.1.11. *Monotonicity of Span*
Let $\mathcal{U}$ and $\mathcal{W}$ be subsets of $\mathcal{V}$. If $\mathcal{U} \subseteq \mathcal{W}$, then $\operatorname{span}(\mathcal{U}) \subseteq \operatorname{span}(\mathcal{W})$.

Proof. If $\mathcal{U} = \emptyset$, then $\operatorname{span}(\mathcal{U}) = \{\mathbf{0}\} \subseteq \operatorname{span}(\mathcal{W})$.
Suppose $\mathcal{U} \neq \emptyset$. Every vector in $\operatorname{span}(\mathcal{U})$ is a finite linear combination of vectors in $\mathcal{U}$; since $\mathcal{U} \subseteq \mathcal{W}$, the same expression is a finite linear combination of vectors in $\mathcal{W}$. Hence $\operatorname{span}(\mathcal{U}) \subseteq \operatorname{span}(\mathcal{W})$.
Structural Commentary: Enlarging the generating set can only enlarge the collection of linear combinations.
Key Insight: Span is monotone with respect to inclusion of generating sets. □

Definition 1.1.1.12. *Linear Independence*
A list $(\mathbf{v}_1, \ldots, \mathbf{v}_m)$ in $\mathcal{V}$ is *linearly independent* if $a_1\mathbf{v}_1 + \cdots + a_m\mathbf{v}_m = \mathbf{0}$ implies $a_1 = \cdots = a_m = 0$. The empty list is linearly independent. Any list that is not linearly independent is linearly dependent.

Proposition 1.1.1.13. *Unique Coefficients for Independent Lists*
Let $(\mathbf{v}_1, \ldots, \mathbf{v}_m)$ be a linearly independent list in $\mathcal{V}$.
Then $a_1\mathbf{v}_1 + \cdots + a_m\mathbf{v}_m = b_1\mathbf{v}_1 + \cdots + b_m\mathbf{v}_m$ if and only if $a_j = b_j$ for each $j = 1, \ldots, m$.

Proof. ($\Rightarrow$) Subtracting the two sides gives $(a_1 - b_1)\mathbf{v}_1 + \cdots + (a_m - b_m)\mathbf{v}_m = \mathbf{0}$.
Linear independence forces $a_j - b_j = 0$ for every j, so $a_j = b_j$.

($\Leftarrow$) If $a_j = b_j$ for every j, then the two linear combinations have identical coefficients and are equal.

Structural Commentary: Linear independence makes coefficients in a linear combination unique.
Key Insight: Independent vectors allow no hidden coefficient cancellation. □

Proposition 1.1.1.14. *Appending to a Dependent List*
Let $(\mathbf{v}_1, \ldots, \mathbf{v}_m)$ be a linearly dependent list in $\mathcal{V}$.
Then for every $\mathbf{v} \in \mathcal{V}$, the list $(\mathbf{v}_1, \ldots, \mathbf{v}_m, \mathbf{v})$ is linearly dependent.

Proof. Since $(\mathbf{v}_1, \ldots, \mathbf{v}_m)$ is linearly dependent, $\exists$ scalars $a_1, \ldots, a_m$, not all zero, s.t. $a_1\mathbf{v}_1 + \cdots + a_m\mathbf{v}_m = \mathbf{0}$.
For any $\mathbf{v} \in \mathcal{V}$, $a_1\mathbf{v}_1 + \cdots + a_m\mathbf{v}_m + 0\mathbf{v} = \mathbf{0}$ and the scalars $a_1, \ldots, a_m, 0$ are not all zero.
Structural Commentary: Longer lists inherit dependence; set new coefficients to 0.
Key Insight: Once a list has a redundancy, adding more vectors cannot remove it. □

Definition 1.1.1.15. *Omitting a Vector from a List*
Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$ be a list in $\mathcal{V}$ with $m \geq 2$.
For $j \in \{1, \ldots, m\}$, the list obtained from β by omitting $\mathbf{v}_j$ is denoted $(\mathbf{v}_1, \ldots, \widehat{\mathbf{v}_j}, \ldots, \mathbf{v}_m)$.

Proposition 1.1.1.16. *Omitting from an Independent List*
If $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$ is linearly independent and $m \geq 2$, then $(\mathbf{v}_1, \ldots, \widehat{\mathbf{v}_j}, \ldots, \mathbf{v}_m)$ is linearly independent for every $j \in \{1, \ldots, m\}$.

Proof. Fix j, and suppose $\sum_{\substack{i=1 \\ i \neq j}}^{m} a_i\mathbf{v}_i = \mathbf{0}$. Set $a_j = 0$.

Then $\sum_{i=1}^{m} a_i\mathbf{v}_i = \mathbf{0}$, so independence of β gives $a_i = 0$ for every $i \neq j$.
Structural Commentary: Shorter lists inherit independence; deletion cannot create cancellation.
Key Insight: Every sublist of an independent list remains independent. □

Proposition 1.1.1.17. *Adjoining and Removing Vectors from Lists*
Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$ be a list in $\mathcal{V}$. Write $(\beta, \mathbf{v}) := (\mathbf{v}_1, \ldots, \mathbf{v}_m, \mathbf{v})$ and $\beta_{\widehat{j}} := (\mathbf{v}_1, \ldots, \widehat{\mathbf{v}_j}, \ldots, \mathbf{v}_m)$.

(i) If β is linearly independent and $\mathbf{v} \notin \operatorname{span}(\beta)$, then $(\beta, \mathbf{v})$ is linearly independent.

(ii) If $\sum_{i=1}^{m} a_i\mathbf{v}_i = \mathbf{0}$ is nontrivial, $a_j \neq 0$, then $\operatorname{span}(\beta_{\widehat{j}}) = \operatorname{span}(\beta)$. Hence $\operatorname{span}(\beta) = \mathcal{V}$ implies $\operatorname{span}(\beta_{\widehat{j}}) = \mathcal{V}$.

Proof. (i) If $\sum_{i=1}^{m} a_i\mathbf{v}_i + a\mathbf{v} = \mathbf{0}$ and $a \neq 0$, then $\mathbf{v} = -a^{-1}\sum_{i=1}^{m} a_i\mathbf{v}_i \in \operatorname{span}(\beta)$, a contradiction.

(ii) Since $a_j \neq 0$, $\mathbf{v}_j = -\sum_{\substack{i=1 \\ i\neq j}}^{m} \frac{a_i}{a_j}\mathbf{v}_i \in \text{span}(\beta_{\widehat{j}})$. Hence $\text{span}(\beta) \subseteq \text{span}(\beta_{\widehat{j}})$; the reverse inclusion is immediate.

Structural Commentary: New directions preserve independence; redundant vectors preserve span.
Key Insight: Add new directions; remove redundant ones. □

Lemma 1.1.1.18. *Replacement Lemma*
Let $\mathcal{V} \neq \{\mathbf{0}\}$, let $\beta = (\mathbf{u}_1, \ldots, \mathbf{u}_r)$ span $\mathcal{V}$, and let $\mathbf{0} \neq \mathbf{v} = \sum_{i=1}^{r} c_i\mathbf{u}_i$.
For $j \in \{1, \ldots, r\}$, set $\beta_{j\leftarrow\mathbf{v}} := (\mathbf{u}_1, \ldots, \mathbf{u}_{j-1}, \mathbf{v}, \mathbf{u}_{j+1}, \ldots, \mathbf{u}_r)$.

(i) $c_j \neq 0$ for some j.
(ii) If $c_j \neq 0$, then $\text{span}(\beta_{j\leftarrow\mathbf{v}}) = \mathcal{V}$; if β is a basis, then $\beta_{j\leftarrow\mathbf{v}}$ is a basis.
(iii) If $r \geq 2$, β is a basis, and $\mathbf{v} \notin \text{span}(\mathbf{u}_1, \ldots, \mathbf{u}_k)$ for some $1 \leq k < r$, then $\beta_{j\leftarrow\mathbf{v}}$ is a basis for some $j > k$.

Proof. (i) If all $c_j = 0$, then $\mathbf{v} = \mathbf{0}$.

(ii) Since $\mathbf{u}_j = c_j^{-1}\left(\mathbf{v} - \sum_{i\neq j} c_i\mathbf{u}_i\right)$, every vector of β lies in $\text{span}(\beta_{j\leftarrow\mathbf{v}})$, while every vector of $\beta_{j\leftarrow\mathbf{v}}$ lies in $\text{span}(\beta)$. Hence $\text{span}(\beta_{j\leftarrow\mathbf{v}}) = \mathcal{V}$. If β is a basis and $\sum_{i<j} b_i\mathbf{u}_i + a\mathbf{v} + \sum_{i>j} b_i\mathbf{u}_i = \mathbf{0}$, then substitution gives $\sum_{i\neq j}(b_i + ac_i)\mathbf{u}_i + ac_j\mathbf{u}_j = \mathbf{0}$. Linear independence of β gives $ac_j = 0$ and $b_i + ac_i = 0$ for $i \neq j$.
Since $c_j \neq 0$, $a = 0$, and then every $b_i = 0$. Thus $\beta_{j\leftarrow\mathbf{v}}$ is also linearly independent.

(iii) If $c_i = 0$ $\forall i > k$, then $\mathbf{v} \in \text{span}(\mathbf{u}_1, \ldots, \mathbf{u}_k)$, contradiction. Thus some $c_j \neq 0$ with $j > k$, and (ii) applies.

Structural Commentary: Replacement preserves span.
Key Insight: Replace where the coefficient is nonzero. □

Lemma 1.1.1.19. *Linear Dependence Lemma*
Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$ be linearly dependent with $\mathbf{v}_1 \neq \mathbf{0}$, and set $\beta_{\widehat{j}} := (\mathbf{v}_1, \ldots, \widehat{\mathbf{v}_j}, \ldots, \mathbf{v}_m)$.
Then $\exists j \in \{2, \ldots, m\}$ such that $\mathbf{v}_j \in \text{span}(\mathbf{v}_1, \ldots, \mathbf{v}_{j-1})$ and $\text{span}(\beta_{\widehat{j}}) = \text{span}(\beta)$.

Proof. Choose a nontrivial relation $\sum_{i=1}^{m} a_i\mathbf{v}_i = \mathbf{0}$ and let $j = \max\{i : a_i \neq 0\}$.
If $j = 1$, then $a_1\mathbf{v}_1 = \mathbf{0}$ with $a_1 \neq 0$, so $\mathbf{v}_1 = \mathbf{0}$, contrary to hypothesis. Hence $j \geq 2$.
Since $a_i = 0$ for $i > j$, we have $\mathbf{v}_j = -\sum_{i=1}^{j-1} \frac{a_i}{a_j}\mathbf{v}_i$, so $\mathbf{v}_j \in \text{span}(\mathbf{v}_1, \ldots, \mathbf{v}_{j-1})$.
Therefore every linear combination of β can be rewritten without $\mathbf{v}_j$, which gives $\text{span}(\beta) \subseteq \text{span}(\beta_{\widehat{j}})$.
The reverse inclusion is immediate because $\beta_{\widehat{j}}$ is a sublist of β.
Structural Commentary: The last nonzero coefficient gives the removable vector.
Key Insight: Dependence gives a later vector removable without losing span. □

Theorem 1.1.1.20. *Independent Lists Are No Longer Than Spanning Lists*
If $\mathcal{V}$ is finite-dimensional, $\alpha = (\mathbf{u}_1, \ldots, \mathbf{u}_m)$ is linearly independent, and $\gamma = (\mathbf{w}_1, \ldots, \mathbf{w}_n)$ spans $\mathcal{V}$, then $m \leq n$.

Proof. Suppose $m > n$. If $n = 0$, then $\text{span}(\gamma) = \{\mathbf{0}\}$, so $\mathcal{V} = \{\mathbf{0}\}$ and no nonempty independent list exists; hence $n \geq 1$. Starting from γ, we construct for each $0 \leq k \leq n$ a spanning list of length n containing $\mathbf{u}_1, \ldots, \mathbf{u}_k$.
The case $k = 0$ is γ. Suppose the list has been constructed for some $k < n$. Write $\mathbf{u}_{k+1}$ as a linear combination of its entries. If all coefficients of entries other than $\mathbf{u}_1, \ldots, \mathbf{u}_k$ were zero, then $\mathbf{u}_{k+1} \in \text{span}(\mathbf{u}_1, \ldots, \mathbf{u}_k)$, contradicting the linear independence of α. Thus some remaining entry has nonzero coefficient; replacing that entry by $\mathbf{u}_{k+1}$ preserves span by Lemma 1.1.1.18(ii). This proves the induction.
For $k = n$, the resulting spanning list has length n and contains $\mathbf{u}_1, \ldots, \mathbf{u}_n$; hence it is exactly these vectors, up to order. Thus $\text{span}(\mathbf{u}_1, \ldots, \mathbf{u}_n) = \mathcal{V}$.
Since $m > n$, this puts $\mathbf{u}_{n+1}$ in $\text{span}(\mathbf{u}_1, \ldots, \mathbf{u}_n)$, contradicting the linear independence of α.
Structural Commentary: Independent length is bounded by spanning length.
Key Insight: Independent lists cannot be longer than spanning lists in finite dimensions. □

Definition 1.1.1.21. *Basis*
A list β in $\mathcal{V}$ is a *basis* of $\mathcal{V}$ if it is linearly independent and $\text{span}(\beta) = \mathcal{V}$.

Proposition 1.1.1.22. *Unique Coordinate Representation*
Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$. Then β is a basis of $\mathcal{V}$ iff each $\mathbf{v} \in \mathcal{V}$ has a unique $(a_1, \ldots, a_n) \in \mathbb{F}^n$ with $\mathbf{v} = \sum_{j=1}^{n} a_j \mathbf{v}_j$.

Proof. ($\Rightarrow$) Spanning gives existence; Proposition 1.1.1.13 gives uniqueness.

($\Leftarrow$) Existence gives $\text{span}(\beta) = \mathcal{V}$. If $\sum_{j=1}^{n} a_j \mathbf{v}_j = \mathbf{0}$, uniqueness of zero representation gives $a_1 = \cdots = a_n = 0$.

Structural Commentary: A basis is exactly existence and uniqueness of coordinates.
Key Insight: Coordinates are basis coefficients. □

Theorem 1.1.1.23. *Reducing Spanning Lists*
Every spanning list in a finite-dimensional vector space contains a sublist that is a basis.

Proof. Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ be a spanning list of $\mathcal{V}$. If β is linearly independent, it is a basis. Otherwise, remove one vector without changing the span: if $\mathbf{v}_1 = \mathbf{0}$, omit $\mathbf{v}_1$; if $\mathbf{v}_1 \neq \mathbf{0}$, Lemma 1.1.1.19 gives a removable $\mathbf{v}_j$. The new list is a sublist of β, still spans $\mathcal{V}$, and has length one less.
Repeat this step while the current spanning list is dependent. Since the length decreases each time, the process stops after at most n removals. At termination the remaining sublist still spans $\mathcal{V}$ and is linearly independent, hence is a basis.
Structural Commentary: A spanning list may contain redundant vectors. Removing only vectors forced by the others preserves span until a basis remains.
Key Insight: Spanning gives enough vectors; reduction deletes exactly the unnecessary ones. □

Theorem 1.1.1.24. *Extending Independent Lists*
Every linearly independent list in a finite-dimensional vector space can be extended to a basis.

Proof. Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$ be linearly independent in $\mathcal{V}$, and choose a spanning list $(\mathbf{w}_1, \ldots, \mathbf{w}_n)$ of $\mathcal{V}$.
Starting with $\mathcal{B} = \beta$, inspect $\mathbf{w}_1, \ldots, \mathbf{w}_n$ in order and append $\mathbf{w}_j$ exactly when $\mathbf{w}_j \notin \text{span}(\mathcal{B})$. Each append preserves linear independence by Proposition 1.1.1.17.
After the inspection, each $\mathbf{w}_j$ lies in $\text{span}(\mathcal{B})$: it was either already there when inspected, or it

was appended to $\mathcal{B}$. Since the $\mathbf{w}_j$ span $\mathcal{V}$, every vector of $\mathcal{V}$ is a linear combination of vectors in $\text{span}(\mathcal{B})$, and hence lies in $\text{span}(\mathcal{B})$. Thus $\mathcal{B}$ is a spanning independent list, so it is a basis.
Structural Commentary: An independent list has no redundancy, but may miss directions. Appending only vectors outside its span fills those directions while preserving independence.
Key Insight: A finite-dimensional independent list becomes a basis once enough new directions are added. □

Corollary 1.1.1.25. *Every finite-dimensional vector space has a basis, and every subspace of a finite-dimensional vector space is finite-dimensional.*

Proof. Since $\mathcal{V}$ is finite-dimensional, it has a finite spanning list; Theorem 1.1.1.23 reduces it to a basis.
Let $\mathcal{U} \subseteq \mathcal{V}$ be a subspace. If $\mathcal{U} = \{\mathbf{0}\}$, then the empty list is a basis of $\mathcal{U}$.
Otherwise, begin with any nonzero vector of $\mathcal{U}$, and repeatedly append a vector of $\mathcal{U}$ outside the span of the current list if such a vector exists. Proposition 1.1.1.17 keeps the list linearly independent. Theorem 1.1.1.20 bounds its length by the length of any spanning list of $\mathcal{V}$, so this finite process must terminate. When it terminates, no vector of $\mathcal{U}$ lies outside the span of the list; hence the list spans $\mathcal{U}$ and is a basis of $\mathcal{U}$.
Structural Commentary: Finite-dimensional spaces have finite coordinate systems, and subspaces cannot contain longer independent lists than the ambient space permits.
Key Insight: Finite-dimensionality passes downward to subspaces. □

Theorem 1.1.1.26. *Basis Length Is Well-Defined*
Any two bases of a finite-dimensional vector space have the same length.

Proof. Let $\beta_1 = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$ and $\beta_2 = (\mathbf{w}_1, \ldots, \mathbf{w}_n)$ be bases of $\mathcal{V}$. Since β_1 is linearly independent and β_2 spans $\mathcal{V}$, Theorem 1.1.1.20 gives $m \leq n$. Reversing the roles of the two bases gives $n \leq m$. Hence $m = n$.
Structural Commentary: Bases are simultaneously independent and spanning, so the comparison theorem traps their lengths in both directions. This makes dimension well-defined.
Key Insight: Dimension is an intrinsic number, not an artefact of a particular basis. □

Definition 1.1.1.27. *Dimension*
The zero vector space has *dimension* 0. If $\mathcal{V}$ has a basis of positive integer length n, then $\mathcal{V}$ is *n-dimensional* and $\dim(\mathcal{V}) = n$. A vector space is *finite-dimensional* if it has finite dimension; otherwise it is *infinite-dimensional*.

Proposition 1.1.1.28. *Dimension Bounds and Basis Criteria*
If $\mathcal{V}$ is finite-dimensional and $\mathcal{U} \subseteq \mathcal{V}$ is a subspace, then $\dim(\mathcal{U}) \leq \dim(\mathcal{V})$, with equality if and only if $\mathcal{U} = \mathcal{V}$. Moreover, a spanning list of length $\dim(\mathcal{V})$ is automatically a basis, and a linearly independent list of length $\dim(\mathcal{V})$ is automatically a basis.

Proof. Let $(\mathbf{u}_1, \ldots, \mathbf{u}_m)$ be a basis of $\mathcal{U}$ and $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ a basis of $\mathcal{V}$.
Theorem 1.1.1.20 gives $m \leq n$, so $\dim(\mathcal{U}) \leq \dim(\mathcal{V})$. If $\mathcal{U} = \mathcal{V}$, equality is clear.
Conversely, if $m = n$, then the basis of $\mathcal{U}$ is an independent list in $\mathcal{V}$ of length $\dim(\mathcal{V})$; extending it to a basis of $\mathcal{V}$ adds no vectors, so it already spans $\mathcal{V}$. Hence $\mathcal{U} = \mathcal{V}$. Let γ be a spanning list of length $n = \dim(\mathcal{V})$.
By Theorem 1.1.1.23, γ contains a sublist that is a basis of $\mathcal{V}$. Any proper sublist has length $< n$, impossible by Theorem 1.1.1.26; hence no vector was removed and γ is itself a basis.
Let α be a linearly independent list of length n. By Theorem 1.1.1.24, α extends to a basis of $\mathcal{V}$. Adding any vector would produce a basis of length $> n$, again contradicting Theorem 1.1.1.26; hence no vector was added and α is itself a basis.

Structural Commentary: Dimension bounds subspaces and detects full-size lists. Equality of dimensions forces a subspace to be the whole space.
Key Insight: In dimension n, n independent or n spanning vectors already form a basis. □

Definition 1.1.1.29. *Sums of Subspaces*
If $\mathcal{U}_1, \ldots, \mathcal{U}_m$ are subspaces of $\mathcal{V}$, define $\mathcal{U}_1 + \cdots + \mathcal{U}_m := \{\mathbf{u}_1 + \cdots + \mathbf{u}_m : \mathbf{u}_j \in \mathcal{U}_j\}$.
In particular, for subspaces $\mathcal{U}, \mathcal{W} \subseteq \mathcal{V}$, $\mathcal{U} + \mathcal{W} = \text{span}(\mathcal{U} \cup \mathcal{W}) = \{\mathbf{u} + \mathbf{w} : \mathbf{u} \in \mathcal{U},\ \mathbf{w} \in \mathcal{W}\}$.
If every vector in this sum has a unique such decomposition, then the sum is direct and we write $\mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_m$.
For two subspaces, $\mathcal{U} + \mathcal{W}$ is direct if and only if $\mathcal{U} \cap \mathcal{W} = \{\mathbf{0}\}$; in this case it is denoted $\mathcal{U} \oplus \mathcal{W}$.

Proposition 1.1.1.30. *Uniqueness in a Direct Sum*
Let $\mathcal{U}$ and $\mathcal{W}$ be subspaces of $\mathcal{V}$ with $\mathcal{U} \cap \mathcal{W} = \{\mathbf{0}\}$. Then each vector in $\mathcal{U} \oplus \mathcal{W}$ has a unique representation $\mathbf{u} + \mathbf{w}$ with $\mathbf{u} \in \mathcal{U}$ and $\mathbf{w} \in \mathcal{W}$.

Proof. Existence follows from the definition of $\mathcal{U}+\mathcal{W}$. For uniqueness, suppose $\mathbf{u}_1+\mathbf{w}_1 = \mathbf{u}_2+\mathbf{w}_2$, $\mathbf{u}_1, \mathbf{u}_2 \in \mathcal{U}$, $\mathbf{w}_1, \mathbf{w}_2 \in \mathcal{W}$. Then $\mathbf{u}_1 - \mathbf{u}_2 = \mathbf{w}_2 - \mathbf{w}_1 \in \mathcal{U}, \mathcal{W}$, hence equals $\mathbf{0}$. Thus $\mathbf{u}_1 = \mathbf{u}_2$ and $\mathbf{w}_1 = \mathbf{w}_2$.
Structural Commentary: Trivial intersection prevents cancellation between summands.
Key Insight: Directness means unique decomposition. □

Proposition 1.1.1.31. *Sum of Subspaces Is the Smallest Containing Subspace*
Suppose $\mathcal{U}_1, \ldots, \mathcal{U}_m$ are subspaces of $\mathcal{V}$. Then $\mathcal{U}_1+\cdots+\mathcal{U}_m$ is the smallest subspace of $\mathcal{V}$ containing $\mathcal{U}_1, \ldots, \mathcal{U}_m$.

Proof. Since $\mathbf{0} \in \mathcal{U}_j$ for each j, we have $\mathbf{0} \in \mathcal{U}_1+\cdots+\mathcal{U}_m$. If $\mathbf{x} = \mathbf{u}_1+\cdots+\mathbf{u}_m$ and $\mathbf{y} = \mathbf{v}_1+\cdots+\mathbf{v}_m$ lie in $\mathcal{U}_1 + \cdots + \mathcal{U}_m$, then for any $c \in \mathbb{F}$, $c\mathbf{x} + \mathbf{y} = (c\mathbf{u}_1 + \mathbf{v}_1) + \cdots + (c\mathbf{u}_m + \mathbf{v}_m) \in \mathcal{U}_1 + \cdots + \mathcal{U}_m$, because $c\mathbf{u}_j + \mathbf{v}_j \in \mathcal{U}_j$ for each j. By Theorem 1.1.1.7, $\mathcal{U}_1 + \cdots + \mathcal{U}_m$ is a subspace of $\mathcal{V}$.
Each $\mathcal{U}_j$ is contained in $\mathcal{U}_1 + \cdots + \mathcal{U}_m$ by taking all other summands to be $\mathbf{0}$. If $\mathcal{W}$ is any subspace of $\mathcal{V}$ containing $\mathcal{U}_1, \ldots, \mathcal{U}_m$, then closure under addition gives $\mathbf{u}_1 + \cdots + \mathbf{u}_m \in \mathcal{W}$ whenever $\mathbf{u}_j \in \mathcal{U}_j$. Hence $\mathcal{U}_1 + \cdots + \mathcal{U}_m \subseteq \mathcal{W}$, so the sum is the smallest such subspace.
Structural Commentary: Sums of subspaces are obtained by closing several subspaces under finite addition. Minimality follows because every containing subspace is already closed under those sums.
Key Insight: The sum adds no extra structure beyond the linear combinations forced by the given subspaces. □

Proposition 1.1.1.32. *Equivalent Conditions for a Direct Sum*
For subspaces $\mathcal{U}_1, \ldots, \mathcal{U}_m \subseteq \mathcal{V}$, the following are equivalent:

i. $\mathcal{V} = \mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_m$;

ii. *every $\mathbf{v} \in \mathcal{V}$ has a unique representation $\mathbf{v} = \mathbf{u}_1 + \cdots + \mathbf{u}_m$ with $\mathbf{u}_j \in \mathcal{U}_j$;*

iii. $\mathcal{V} = \mathcal{U}_1 + \cdots + \mathcal{U}_m$, *and if $\mathbf{u}_1 + \cdots + \mathbf{u}_m = \mathbf{0}$ with $\mathbf{u}_j \in \mathcal{U}_j$, then $\mathbf{u}_1 = \cdots = \mathbf{u}_m = \mathbf{0}$.*

Proof. $(i) \Rightarrow (ii)$ This follows from the definition of direct sum.

$(ii) \Rightarrow (iii)$ Every vector lies in $\mathcal{U}_1 + \cdots + \mathcal{U}_m$, so $\mathcal{V} = \mathcal{U}_1 + \cdots + \mathcal{U}_m$. If $\mathbf{u}_1 + \cdots + \mathbf{u}_m = \mathbf{0}$, then this representation of $\mathbf{0}$ must equal $\mathbf{0} + \cdots + \mathbf{0}$ by uniqueness, so each $\mathbf{u}_j = \mathbf{0}$.

$(iii) \Rightarrow (i)$ Since $\mathcal{V} = \mathcal{U}_1 + \cdots + \mathcal{U}_m$, every $\mathbf{v} \in \mathcal{V}$ has at least one representation. If $\mathbf{v} = \mathbf{u}_1+\cdots+\mathbf{u}_m = \mathbf{w}_1+\cdots+\mathbf{w}_m$, then $(\mathbf{u}_1-\mathbf{w}_1)+\cdots+(\mathbf{u}_m-\mathbf{w}_m) = \mathbf{0}$. By iii., $\mathbf{u}_j - \mathbf{w}_j = \mathbf{0}$ for every j, so the representation is unique. Hence $\mathcal{V} = \mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_m$.

Structural Commentary: Direct sums require both spanning by the summands and no cancellation among them. The zero-representation condition isolates the no-cancellation part.
Key Insight: Uniqueness is equivalent to saying the only way the summands add to zero is trivially. □

Definition 1.1.1.33. *Product of Vector Spaces*
Let $\mathcal{V}_1, \ldots, \mathcal{V}_m$ be vector spaces over $\mathbb{F}$. Their *product* is $\mathcal{V}_1 \times \cdots \times \mathcal{V}_m := \{(\mathbf{v}_1, \ldots, \mathbf{v}_m) : \mathbf{v}_j \in \mathcal{V}_j\}$. Addition, scalar multiplication defined componentwise: $(\mathbf{u}_1, \ldots, \mathbf{u}_m) + (\mathbf{v}_1, \ldots, \mathbf{v}_m) := (\mathbf{u}_1 + \mathbf{v}_1, \ldots, \mathbf{u}_m + \mathbf{v}_m)$ and $a(\mathbf{v}_1, \ldots, \mathbf{v}_m) := (a\mathbf{v}_1, \ldots, a\mathbf{v}_m)$.

Proposition 1.1.1.34. *Products Are Vector Spaces*
If $\mathcal{V}_1, \ldots, \mathcal{V}_m$ are vector spaces over $\mathbb{F}$, then $\mathcal{V}_1 \times \cdots \times \mathcal{V}_m$ is a vector space over $\mathbb{F}$ under componentwise operations.

Proof. The additive identity is $(\mathbf{0}_1, \ldots, \mathbf{0}_m)$, where $\mathbf{0}_j$ is the zero vector of $\mathcal{V}_j$, and additive inverse of $(\mathbf{v}_1, \ldots, \mathbf{v}_m)$ is $(-\mathbf{v}_1, \ldots, -\mathbf{v}_m)$. All remaining vector space axioms hold componentwise because they hold in each $\mathcal{V}_j$.
Structural Commentary: Product spaces collect several vector spaces into one coordinatewise structure.
Key Insight: Every vector space axiom reduces to the same axiom in each component. □

Theorem 1.1.1.35. *Dimension of a Product*
If $\mathcal{V}_1, \ldots, \mathcal{V}_m$ are finite-dimensional vector spaces over $\mathbb{F}$, then $\mathcal{V}_1 \times \cdots \times \mathcal{V}_m$ is finite-dimensional and

$$\dim(\mathcal{V}_1 \times \cdots \times \mathcal{V}_m) = \dim(\mathcal{V}_1) + \cdots + \dim(\mathcal{V}_m)$$

Proof. For each j, let $(\mathbf{v}_{j,1}, \ldots, \mathbf{v}_{j,n_j})$ be a basis of $\mathcal{V}_j$. For every j and k, let $\mathbf{e}_{j,k}$ denote the element of $\mathcal{V}_1 \times \cdots \times \mathcal{V}_m$ whose jth component is $\mathbf{v}_{j,k}$ and whose other components are zero.
The list of all $\mathbf{e}_{j,k}$ spans the product because each component can be expanded in its own basis. If a linear combination of the $\mathbf{e}_{j,k}$ is zero, then looking at the jth component gives a linear relation among $(\mathbf{v}_{j,1}, \ldots, \mathbf{v}_{j,n_j})$, so all coefficients in that component are zero. Hence the list is linearly independent and therefore is a basis of the product. Its length is $n_1 + \cdots + n_m$.
Structural Commentary: A basis for a product is formed by placing each component basis into its own slot.
Key Insight: Product dimension counts the independent directions contributed by each factor. □

Theorem 1.1.1.36. *Dimension Formula for a Sum*
If $\mathcal{U}, \mathcal{W}$ are finite-dimensional subspaces of $\mathcal{V}$, then $\dim(\mathcal{U}+\mathcal{W}) = \dim(\mathcal{U})+\dim(\mathcal{W})-\dim(\mathcal{U}\cap\mathcal{W})$.

Proof. Choose a basis $(\mathbf{e}_1, \ldots, \mathbf{e}_r)$ of $\mathcal{U} \cap \mathcal{W}$, extend it to a basis $(\mathbf{e}_1, \ldots, \mathbf{e}_r, \mathbf{u}_1, \ldots, \mathbf{u}_p)$ of $\mathcal{U}$, and to a basis $(\mathbf{e}_1, \ldots, \mathbf{e}_r, \mathbf{w}_1, \ldots, \mathbf{w}_q)$ of $\mathcal{W}$. Then $(\mathbf{e}_1, \ldots, \mathbf{e}_r, \mathbf{u}_1, \ldots, \mathbf{u}_p, \mathbf{w}_1, \ldots, \mathbf{w}_q)$ spans $\mathcal{U} + \mathcal{W}$ because every vector in the sum is the sum of one vector from $\mathcal{U}$ and one vector from $\mathcal{W}$.
To prove linear independence, suppose $a_1\mathbf{e}_1 + \cdots + a_r\mathbf{e}_r + b_1\mathbf{u}_1 + \cdots + b_p\mathbf{u}_p + c_1\mathbf{w}_1 + \cdots + c_q\mathbf{w}_q = \mathbf{0}$. Then $a_1\mathbf{e}_1 + \cdots + a_r\mathbf{e}_r + b_1\mathbf{u}_1 + \cdots + b_p\mathbf{u}_p$ lies in $\mathcal{U} \cap \mathcal{W}$. Since $(\mathbf{e}_1, \ldots, \mathbf{e}_r)$ is a basis of $\mathcal{U} \cap \mathcal{W}$ and $(\mathbf{e}_1, \ldots, \mathbf{e}_r, \mathbf{u}_1, \ldots, \mathbf{u}_p)$ is a basis of $\mathcal{U}$, we get $b_1 = \cdots = b_p = 0$. The remaining relation lies in the basis $(\mathbf{e}_1, \ldots, \mathbf{e}_r, \mathbf{w}_1, \ldots, \mathbf{w}_q)$ of $\mathcal{W}$, so $a_1 = \cdots = a_r = c_1 = \cdots = c_q = 0$. Thus the displayed list is a basis of $\mathcal{U} + \mathcal{W}$.
Therefore $\dim(\mathcal{U} + \mathcal{W}) = r + p + q = (r + p) + (r + q) - r$.
Structural Commentary: The formula corrects double-counting: vectors in $\mathcal{U} \cap \mathcal{W}$ are counted once in each subspace. Extending an intersection basis separates shared directions from new

directions.
Key Insight: The dimension of a sum is the sum of dimensions minus the overlap. □

Corollary 1.1.1.37. *Dimension of a Direct Sum*
If $\mathcal{U}$ and $\mathcal{W}$ are finite-dimensional subspaces of $\mathcal{V}$ with $\mathcal{U} \cap \mathcal{W} = \{\mathbf{0}\}$, then $\dim(\mathcal{U} \oplus \mathcal{W}) = \dim(\mathcal{U}) + \dim(\mathcal{W})$.

Proof. Since $\mathcal{U} \cap \mathcal{W} = \{\mathbf{0}\}$, the sum is direct and $\dim(\mathcal{U} \cap \mathcal{W}) = 0$. The result follows from Theorem 1.1.1.36.
Structural Commentary: When the overlap is trivial, there is no double-counting in the dimension formula. Direct sums are exactly the case where the overlap term vanishes.
Key Insight: Dimensions add exactly for direct sums. □

Theorem 1.1.1.38. *Direct Sums and Dimension*
Suppose $\mathcal{V}$ is finite-dimensional and $\mathcal{U}_1, \ldots, \mathcal{U}_m$ are subspaces of $\mathcal{V}$. Then $\mathcal{U}_1 + \cdots + \mathcal{U}_m$ is a direct sum if and only if $\dim(\mathcal{U}_1 + \cdots + \mathcal{U}_m) = \dim(\mathcal{U}_1) + \cdots + \dim(\mathcal{U}_m)$.

Proof. For each j, choose a basis β_j of $\mathcal{U}_j$, and concatenate these bases into a list β.
This list spans $\mathcal{U}_1 + \cdots + \mathcal{U}_m$ and has length $\dim(\mathcal{U}_1) + \cdots + \dim(\mathcal{U}_m)$.

- ($\Rightarrow$) If the sum is direct, any linear relation among the vectors in β gives a zero decomposition $\mathbf{u}_1 + \cdots + \mathbf{u}_m = \mathbf{0}$ with $\mathbf{u}_j \in \mathcal{U}_j$. Directness gives $\mathbf{u}_j = \mathbf{0}$ for every j, and independence of each β_j gives all coefficients zero. Thus β is a basis of the sum, and the dimensions add.
- ($\Leftarrow$) If the dimensions add, then β is a spanning list whose length equals the dimension of the sum, so Proposition 1.1.1.28 makes β a basis. A zero decomposition $\mathbf{u}_1 + \cdots + \mathbf{u}_m = \mathbf{0}$ expands to a zero linear relation in this basis, so every $\mathbf{u}_j = \mathbf{0}$. Hence the sum is direct.

Structural Commentary: Direct sums add dimensions without hidden overlap.
Key Insight: Equality of dimensions means the concatenated bases have no linear redundancy. □

Corollary 1.1.1.39. *Large Subspaces Must Intersect*
Let $\mathcal{U}$ and $\mathcal{W}$ be subspaces of a finite-dimensional vector space $\mathcal{V}$, and let k be a positive integer.

i. If $\dim(\mathcal{U}) + \dim(\mathcal{W}) > \dim(\mathcal{V})$, *then $\mathcal{U} \cap \mathcal{W}$ contains a nonzero vector.*

ii. If $\dim(\mathcal{U}) + \dim(\mathcal{W}) \geq \dim(\mathcal{V}) + k$, *then $\mathcal{U} \cap \mathcal{W}$ contains k linearly independent vectors.*

Proof. Since $\mathcal{U} + \mathcal{W} \subseteq \mathcal{V}$, Theorem 1.1.1.36 gives

$$\dim(\mathcal{U} \cap \mathcal{W}) = \dim(\mathcal{U}) + \dim(\mathcal{W}) - \dim(\mathcal{U} + \mathcal{W}) \geq \dim(\mathcal{U}) + \dim(\mathcal{W}) - \dim(\mathcal{V})$$

If $\dim(\mathcal{U}) + \dim(\mathcal{W}) > \dim(\mathcal{V})$, the right-hand side is at least 1, so the intersection contains a nonzero vector. If $\dim(\mathcal{U}) + \dim(\mathcal{W}) \geq \dim(\mathcal{V}) + k$, then $\dim(\mathcal{U} \cap \mathcal{W}) \geq k$, so intersection contains k linearly independent vectors.
Structural Commentary: Excess dimension appears as overlap.
Key Insight: Two large subspaces cannot fit inside $\mathcal{V}$ without sharing directions. □

Proposition 1.1.1.40. *Complements Exist in Finite Dimensions*
If $\mathcal{V}$ is finite-dimensional and $\mathcal{U} \subseteq \mathcal{V}$ is a subspace, then there exists a subspace $\mathcal{W}$ such that $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$.

Proof. Choose a basis $(\mathbf{u}_1, \ldots, \mathbf{u}_m)$ of $\mathcal{U}$ and extend it to a basis $(\mathbf{u}_1, \ldots, \mathbf{u}_m, \mathbf{w}_1, \ldots, \mathbf{w}_n)$ of $\mathcal{V}$. Let $\mathcal{W} = \operatorname{span}(\mathbf{w}_1, \ldots, \mathbf{w}_n)$. Every vector in $\mathcal{V}$ is a linear combination of this extended basis, so $\mathcal{V} = \mathcal{U} + \mathcal{W}$. If $\mathbf{v} \in \mathcal{U} \cap \mathcal{W}$, then $\mathbf{v}$ has one representation using only $\mathbf{u}_1, \ldots, \mathbf{u}_m$ and another using only $\mathbf{w}_1, \ldots, \mathbf{w}_n$. Uniqueness of coefficients in the basis of $\mathcal{V}$ forces $\mathbf{v} = \mathbf{0}$. Hence $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$.
Structural Commentary: Complements split a space into a subspace and the remaining basis directions.
Key Insight: Extend a basis of $\mathcal{U}$ to split $\mathcal{V}$. □

Definition 1.1.1.41. *Affine Subset and Parallel Subspaces*
Let $\mathcal{U}$ be a subspace of $\mathcal{V}$. For $\mathbf{v} \in \mathcal{V}$, the subset $\mathbf{v} + \mathcal{U} := \{\mathbf{v} + \mathbf{u} : \mathbf{u} \in \mathcal{U}\}$ is an *affine subset* of $\mathcal{V}$ parallel to $\mathcal{U}$.

Proposition 1.1.1.42. *Parallel Affine Subsets Are Equal or Disjoint*
Let $\mathcal{U}$ be a subspace of $\mathcal{V}$ and let $\mathbf{v}, \mathbf{w} \in \mathcal{V}$. The following are equivalent:

(i) $\mathbf{v} - \mathbf{w} \in \mathcal{U}$;

(ii) $\mathbf{v} + \mathcal{U} = \mathbf{w} + \mathcal{U}$;

(iii) $(\mathbf{v} + \mathcal{U}) \cap (\mathbf{w} + \mathcal{U}) \neq \emptyset$.

Proof. *(i)* $\Rightarrow$ *(ii)* If $\mathbf{v} - \mathbf{w} \in \mathcal{U}$ and $\mathbf{u} \in \mathcal{U}$, then $\mathbf{v} + \mathbf{u} = \mathbf{w} + ((\mathbf{v} - \mathbf{w}) + \mathbf{u}) \in \mathbf{w} + \mathcal{U}$. Thus $\mathbf{v} + \mathcal{U} \subseteq \mathbf{w} + \mathcal{U}$, and the reverse inclusion is similar.

(ii) $\Rightarrow$ *(iii)* Equal affine subsets have the same elements.

(iii) $\Rightarrow$ *(i)* If $\mathbf{x} \in (\mathbf{v} + \mathcal{U}) \cap (\mathbf{w} + \mathcal{U})$, then $\mathbf{x} = \mathbf{v} + \mathbf{u}_1 = \mathbf{w} + \mathbf{u}_2$ for some $\mathbf{u}_1, \mathbf{u}_2 \in \mathcal{U}$. Hence $\mathbf{v} - \mathbf{w} = \mathbf{u}_2 - \mathbf{u}_1 \in \mathcal{U}$.

Structural Commentary: Parallel affine subsets either coincide or do not meet.
Key Insight: Intersection is controlled by the difference of base points. □

Definition 1.1.1.43. *Quotient Space*
Let $\mathcal{U}$ be a subspace of $\mathcal{V}$. The *quotient space* $\mathcal{V}/\mathcal{U}$ is the set of all affine subsets of $\mathcal{V}$ parallel to $\mathcal{U}$:

$$\mathcal{V}/\mathcal{U} := \{\mathbf{v} + \mathcal{U} : \mathbf{v} \in \mathcal{V}\}.$$

Proposition 1.1.1.44. *Structure of the Quotient*
Let $\mathcal{U}$ be a subspace of $\mathcal{V}$. Addition and scalar multiplication on $\mathcal{V}/\mathcal{U}$ defined by $(\mathbf{v} + \mathcal{U}) + (\mathbf{w} + \mathcal{U}) := (\mathbf{v} + \mathbf{w}) + \mathcal{U}$ and $a(\mathbf{v} + \mathcal{U}) := a\mathbf{v} + \mathcal{U}$ are well-defined, and with these operations $\mathcal{V}/\mathcal{U}$ is a vector space.

Proof. If $\mathbf{v} + \mathcal{U} = \mathbf{v}' + \mathcal{U}$ and $\mathbf{w} + \mathcal{U} = \mathbf{w}' + \mathcal{U}$, then $\mathbf{v} - \mathbf{v}' \in \mathcal{U}$ and $\mathbf{w} - \mathbf{w}' \in \mathcal{U}$.
Since $\mathcal{U}$ is a subspace, $(\mathbf{v} + \mathbf{w}) - (\mathbf{v}' + \mathbf{w}') \in \mathcal{U}$, and $a\mathbf{v} - a\mathbf{v}' \in \mathcal{U}$ so the operations do not depend on the chosen representatives. The vector space axioms are inherited from $\mathcal{V}$ under these operations.
Structural Commentary: Quotients preserve operations modulo $\mathcal{U}$.
Key Insight: Cosets identify vectors differing by elements of $\mathcal{U}$. □

Definition 1.1.1.45. *Quotient Map*
Let $\mathcal{U}$ be a subspace of $\mathcal{V}$. The *quotient map* is the map $\pi : \mathcal{V} \to \mathcal{V}/\mathcal{U}$ defined by $\pi(\mathbf{v}) = \mathbf{v} + \mathcal{U}$.

Proposition 1.1.1.46. *Properties of the Quotient Map*
The quotient map π is linear, surjective, and $\ker(\pi) = \mathcal{U}$.

Proof. For $\mathbf{v}, \mathbf{w} \in \mathcal{V}$ and $a \in \mathbb{F}$, $\pi(a\mathbf{v}+\mathbf{w}) = (a\mathbf{v}+\mathbf{w})+\mathcal{U} = a(\mathbf{v}+\mathcal{U})+(\mathbf{w}+\mathcal{U}) = a\pi(\mathbf{v})+\pi(\mathbf{w})$ so π is linear. It is surjective because every coset in $\mathcal{V}/\mathcal{U}$ has the form $\mathbf{v}+\mathcal{U} = \pi(\mathbf{v})$. Finally, $\ker(\pi) = \{\mathbf{v} \in \mathcal{V} : \mathbf{v}+\mathcal{U} = \mathcal{U}\} = \mathcal{U}$.
Structural Commentary: The quotient map collapses $\mathcal{U}$ to zero and records cosets.
Key Insight: Quotient maps realise subspaces as kernels. □

Theorem 1.1.1.47. *Dimension of a Quotient*
If $\mathcal{V}$ is finite-dimensional and $\mathcal{U}$ is a subspace, then $\dim(\mathcal{V}/\mathcal{U}) = \dim(\mathcal{V}) - \dim(\mathcal{U})$.

Proof. Choose a basis $(\mathbf{u}_1, \ldots, \mathbf{u}_m)$ of $\mathcal{U}$ and extend it to a basis $(\mathbf{u}_1, \ldots, \mathbf{u}_m, \mathbf{w}_1, \ldots, \mathbf{w}_k)$ of $\mathcal{V}$. The cosets $\mathbf{w}_1+\mathcal{U}, \ldots, \mathbf{w}_k+\mathcal{U}$ span $\mathcal{V}/\mathcal{U}$ because the $\mathbf{u}_j$-terms vanish in the quotient.
If $a_1(\mathbf{w}_1+\mathcal{U}) + \cdots + a_k(\mathbf{w}_k+\mathcal{U}) = \mathcal{U}$, then $a_1\mathbf{w}_1 + \cdots + a_k\mathbf{w}_k \in \mathcal{U}$, so a linear relation in the extended basis forces $a_1 = \cdots = a_k = 0$. Hence these cosets form a basis of $\mathcal{V}/\mathcal{U}$, and its length is $k = \dim(\mathcal{V}) - \dim(\mathcal{U})$.
Structural Commentary: Quotiening removes the directions in $\mathcal{U}$ and keeps the added basis directions.
Key Insight: Passing to $\mathcal{V}/\mathcal{U}$ subtracts the collapsed dimension. □

This section develops two complementary ways to organise a vector space: build from smaller pieces by spans and direct sums, or collapse a chosen subspace by passing to a quotient. Bases and dimension measure exactly what is added, split off, or made invisible.